05/02/2025

Code-A

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 | NBTS for NEET-2024-25 (OYMP1)-Test-06 | Time : 180 Min.

Topics Covered:
Complete Updated Syllabus of NEET-2025

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

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1. An ammeter of resistance 20 Ω measures upto 50 mA. The value of shunt required in parallel to measure current upto 5 A is nearly

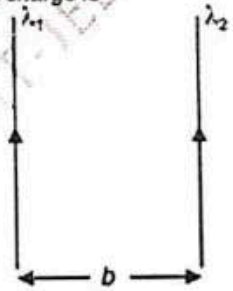
(1) 0.1 Ω

(2) 0.2 Ω

(3) 0.02 Ω

(4) 0.01 Ω

2. Two parallel infinite line charges of linear charge density λ_1 and λ_2 are placed at a distance b apart as shown in the figure. The force acting on a unit length of each linear charge is



(1) $\frac{\lambda_1\lambda_2}{4\pi\epsilon_0 b}$

(2) $\frac{\lambda_1\lambda_2}{2\pi\epsilon_0 b}$

(3) $\frac{\lambda_1\lambda_2}{\pi\epsilon_0 b}$

(4) $\frac{2\lambda_1\lambda_2}{\pi\epsilon_0 b}$

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3. Ratio of the speed of the electron in the ground state of hydrogen to the speed of light in vacuum is

(1) $\frac{1}{337}$

(2) $\frac{1}{137}$

(3) $\frac{2}{337}$

(4) $\frac{1}{2}$

4. When intensities of light coming from two slits in a Y.D.S.E. have ratio $\sqrt{2} : \sqrt{3}$, then the ratio of slit widths is

(1) 2 : 3

(2) 4 : 9

(3) $3 : \sqrt{2}$

(4) $\sqrt{2} : \sqrt{3}$

5. A short linear dipole of dipole moment $\vec{P}$ is kept in electric field $\vec{E}$ such that, $\vec{E}$ and $\vec{P}$ are in the opposite directions. The magnitude of net torque acting on the dipole is

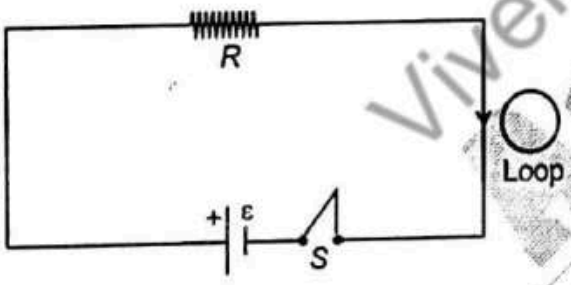
(1) PE

(2) Zero

(3) $\frac{PE}{2}$

(4) $\frac{\sqrt{3}PE}{2}$

6. The switch S in the circuit below is closed first and then opened. The closed conducting loop will show



(1) An anticlockwise current

(2) A clockwise current

(3) An anticlockwise current and then clockwise current

(4) A clockwise current and then anticlockwise current

7. A conducting sphere of radius R is given a charge Q. The electric potential and the electric field at the centre of the sphere respectively are

(1) Both are zero

(2) $\frac{Q}{4\pi\varepsilon_0 R}$ and zero

(3) Zero and $\frac{Q}{4\pi\varepsilon_0 R^2}$

(4) $\frac{Q}{4\pi\varepsilon_0 R}$ and $\frac{Q}{4\pi\varepsilon_0 R^2}$

8. The work function of a metal is 2 eV. Its threshold wavelength is about

(1) 620 nm

(2) 1240 nm

(3) 310 nm

(4) 900 nm

9. The ratio of average translational kinetic energy to rotational kinetic energy of a diatomic molecule at temperature T is

(1) 3 : 1

(2) 7 : 5

(3) 3 : 2

(4) 5 : 3

10. If nuclear density of ${}_8O^{16}$ is 2.35×10^{17} kg m^{-3}, then nuclear density of element having atomic mass number 32 is

(1) 2.35×10^{17} kg m^{-3}

(2) 4.7×10^{17} kg m^{-3}

(3) 2.35×10^{16} kg m^{-3}

(4) $\frac{2.35 \times 2}{(2)^3} \times 10^{17}$ kgm^{-3}

11. A : Zener breakdown occurs in highly doped semiconductor.
R : Width of depletion layer increases as doping level is increased.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

12. The refracting angle of a prism is A, and refractive index of the material of the prism is $\cot(A/2)$. The angle of minimum deviation is

(1) $180° + 2A$

(2) $180° - 3A$

(3) $180° - 2A$

(4) $90° - A$

13. When water is heated from 0°C to 10°C, its volume

(1) Increases

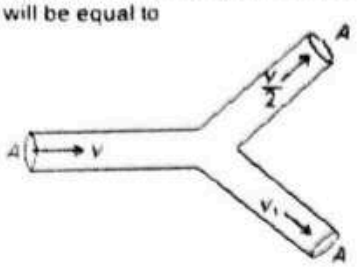
(2) Decreases

(3) First decreases and then increases

(4) Does not change

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14. A tube is shown in figure. If liquid is ideal then velocity v_1 will be equal to



(1) v

(2) Zero

(3) $\frac{v}{3}$

(4) $\frac{v}{2}$

15. An electron is moving in a circular path under the influence of a transverse magnetic field of 3.57×10^{-2} T. If the value of e/m is 1.76×10^{11} C/kg, the frequency of revolution of the electron is

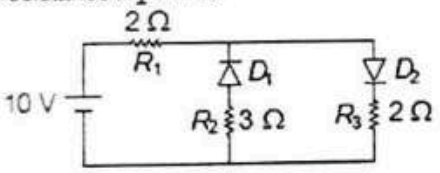
(1) 1 GHz

(2) 100 MHz

(3) 62.8 MHz

(4) 6.28 MHz

16. The given circuit has two ideal diodes connected as shown in the figure below. The current flowing through the resistance R_1 will be



(1) 2.5 A

(2) 10.0 A

(3) 1.43 A

(4) 3.13 A

17. Which of the following is not a characteristic of diamagnetism?

(1) The diamagnetic materials are repelled by a magnetic field

(2) Susceptibility of diamagnetic materials is inversely proportional to the absolute temperature

(3) Magnetic susceptibility of diamagnetic material is small and negative

(4) Diamagnetic material moves from a region of strong magnetic field to weak magnetic field

18. A : Critical angle offered by glass for red light is more than that for blue light.
R : Wavelength of red light is less as compared to blue light.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

19. In which of the following circuits, the maximum power dissipation is observed?

(1) Pure capacitive only

(2) Pure inductive circuit only

(3) Pure resistive circuit

(4) Inductive-capacitive series circuit

20. In a vernier calliper, 10th division on vernier scale coincides with 9th main scale division. If 1 MSD is 0.1 mm, then vernier constant is

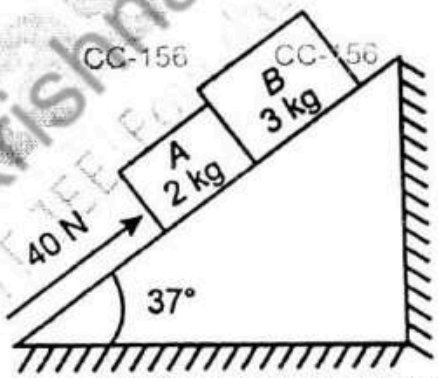
(1) 0.01 cm

(2) 0.001 cm

(3) 0.001 mm

(4) 0.2 mm

21.



A force of 40 N is applied to block A as shown in the figure. The mass of blocks A and B are 2 kg and 3 kg respectively. The blocks slide over a frictionless inclined plane. The force exerted by block A on block B is [given $g = 10\ m\ s^{-2}$]

(1) 18 N

(2) 24 N

(3) 20 N

(4) 32 N

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22. A bob of mass m is whirled in a horizontal plane by means of a string with initial speed ω rad s^{-1}. The tension in string is T. If speed of the bob becomes $\frac{\omega}{2}$ and radius becomes $2r$, then tension in the string becomes

(1) T

(2) $2T$

(3) $\frac{T}{2}$

(4) $\frac{T}{4}$

23. The moment of inertia of thin rod about an axis passing through its mid-point and perpendicular to the rod is 2400 g cm^2. If the mass of the rod is 400 g, then its radius of gyration about an axis passing through its end and perpendicular to rod is

(1) $\sqrt{18}$ cm

(2) $\sqrt{24}$ cm

(3) $\sqrt{72}$ cm

(4) $\sqrt{6}$ cm

24. The particle is projected at angle θ with horizontal ground with speed v, then magnitude of its tangential acceleration at the point of projection is

(1) g cos θ

(2) g sin θ

(3) g

(4) Zero

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25. A particle of mass 1 kg moves along x axis. If position (x) from the origin varies as, $x = t^3$ where x is in meter and t is in seconds. The power delivered by force at $t = 2$ s is

(1) 72 W

(2) 96 W

(3) 120 W

(4) 144 W

26. If a quantity X = (pressure) × (volume), then the dimensions of X is same as that of

(1) Angular momentum

(2) Linear Impulse

(3) Moment of Inertia

(4) Moment of force

27. The Young's modulus of the material of wire is 2×10^{11} N m^{-2}. If the longitudinal strain is 0.1%, then stress in the wire is

(1) 2×10^8 N/m^2

(2) 2×10^7 N/m^2

(3) 10^8 N/m^2

(4) 2×10^9 N/m^2

28. A body of mass m moving with velocity v collides head on and elastically with another body of mass m at rest. The loss in kinetic energy of the bodies in the collision will be

(1) Zero

(2) $\frac{3}{4}mv^2$

(3) $\frac{1}{4}mv^2$

(4) $\frac{1}{2}mv^2$

29. In a spring-block system, the block is oscillating on a horizontal smooth surface with amplitude A and time period T. When the block is at mean position, another identical block of same mass is placed gently over it. Due to this, its

(1) Amplitude of oscillation decreases while time period of oscillation increases

(2) Amplitude of oscillation increase while time period of oscillation decreases

(3) Amplitude of oscillation and Time period, both increase

(4) Amplitude of oscillation and Time period, both decreases

30. A rectangular film of liquid is extended from 4 cm × 2 cm to 4 cm × 4 cm. If the surface tension is 0.250 N m^{-1} for the liquid, the work done is

(1) 0.3 mJ

(2) 0.4 mJ

(3) 0.2 mJ

(4) 0.8 mJ

31. The force per unit charge is equal to

(1) Electric flux

(2) Electric field

(3) Electric potential

(4) Electric current

32. A parallel plate capacitor C is charged to voltage V and then disconnected. The energy stored in the capacitor is U and charge on capacitor is Q. Now the distance between plates is increased to 2 times and the space between them is filled with dielectric of constant $K = 4$. Match the column I and Column II.

Column I	Column II
(A) Charge	(i) Remain unchanged
(B) Energy	(ii) Increase
(C) Capacitance	(iii) Decreases

(1) (A)(i), B(iii), (C)(ii)

(2) (A)(i), (B)(ii), (C)(iii)

(3) (A)(ii), (B)(i), (C)(iii)

(4) (A)(iii), (B)(ii), (C)(i)

33. The maximum energy that can be drawn from the battery of emf 10 V and internal resistance 1 Ω in 10 s

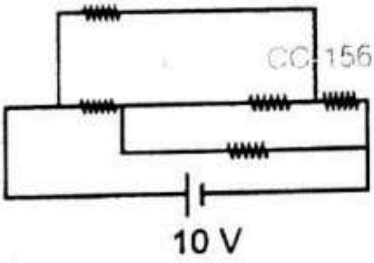
(1) 10 J

(2) 100 J

(3) 1000 J

(4) 10000 J

34. A wire of length l and resistance 50 Ω is divided into 5 equal parts and Joined with the battery as shown. The current supplied by battery is



(1) 1 A

(2) $\frac{1}{2}A$

(3) 2A

(4) $\frac{1}{4}A$

35. A solenoid has a core of a material with relative permeability 100. The windings of solenoid are insulated and carry a current of 2 A. If the number of turns per unit length are 200 per meter, then magnetic field is

(1) 1 T

(2) 0.5 T

(3) 0.05 T

(4) 50 T

36. The magnetic flux through a loop of resistance R changes as $\Delta\phi$ in time Δt then SI unit of $\frac{\Delta\phi}{R}$ is (symbol have their usual meaning)

(1) A s

(2) $A\,s^{-1}$

(3) $J\,V^{-1}$

(4) Both (1) and (3)

37. A circuit has a resistance of 100 Ω and impedance of $100\sqrt{2}$ Ω. Find the reactance of the circuit.

(1) 50 Ω

(2) 20 Ω

(3) 100 Ω

(4) 10 Ω

38. For a light ray undergoing minimum deviation through an equilateral prism

(1) Incident ray will always be parallel to the base of prism

(2) Refracted ray inside prism will always be parallel to the base of prism

(3) Emergent ray will always be parallel to the base of prism

(4) None of these

39. Intrinsic semiconductor is electrically neutral. Extrinsic semiconductor having large number of charge carriers, would be

(1) Positively charged

(2) Negatively charged

(3) Positively charged or negatively depending upon type of impurity added

(4) Electrically neutral

40. The velocity of A and B are $\vec{v_A} = \left(2\hat{i} + 3\hat{j}\right)$ m/s and $\vec{v_B} = \left(3\hat{i} + 4\hat{j}\right)$ m/s. Velocity of B as observed by A is ?

(1) $\left(-\hat{i} + \hat{j}\right)$ m/s

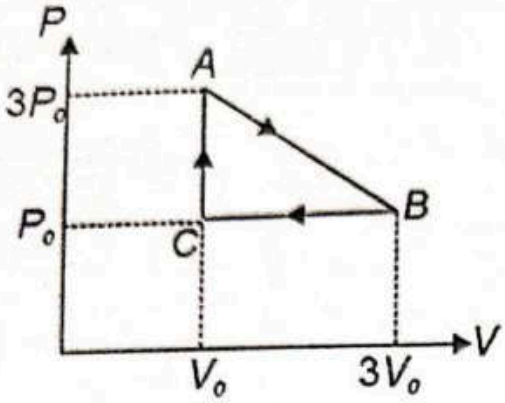
(2) $\left(\hat{i} + \hat{j}\right)$ m/s

(3) $\left(2\hat{i} + 3\hat{j}\right)$ m/s

(4) $\left(-\hat{i} - \hat{j}\right)$ m/s

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41. Work done during the process AB in the P-V diagram shown below is



(1) $2P_0V_0$

(2) $4\ P_0V_0$

(3) $5\ P_0V_0$

(4) $\frac{7}{2}P_0V_0$

42. The variation of energy (E) of particle with position (x) is given by, $E = Ax^2 + Bx$. The factor which is dimensionless if A and B are constant, is

(1) ABx

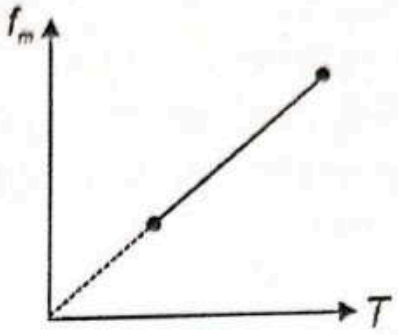
(2) $\frac{AB}{x}$

(3) $\frac{B}{A}x$

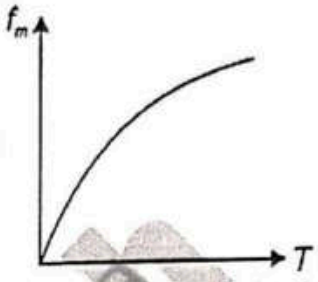
(4) $\frac{Ax}{B}$

43. If f_m is the frequency, corresponding to which the radiant intensity of a block is at its maximum and its absolute temperature is T, then which of the following graph correctly represents the variation of f_m Vs T.

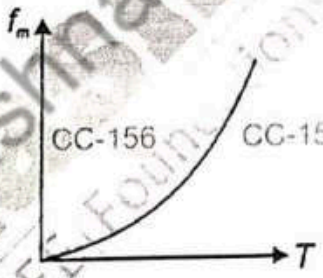
(1)



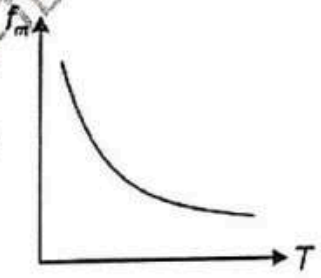
(2)



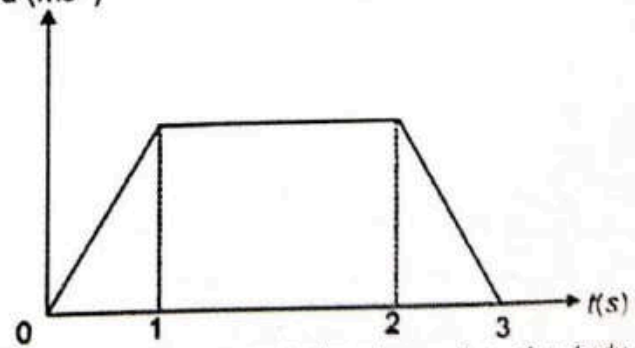
(3)



(4)



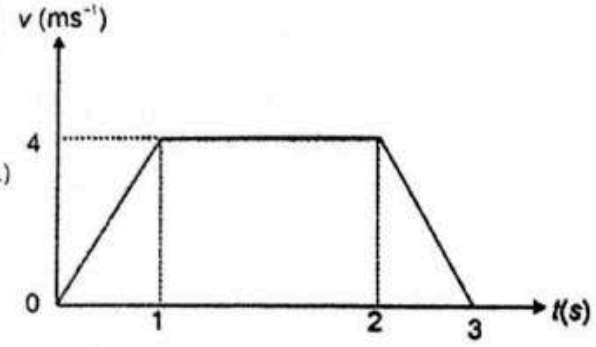
44. $a\ (ms^{-2})$

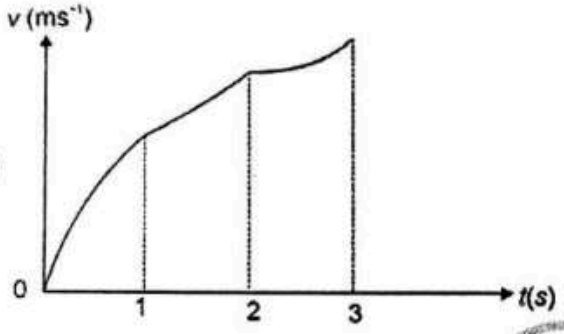


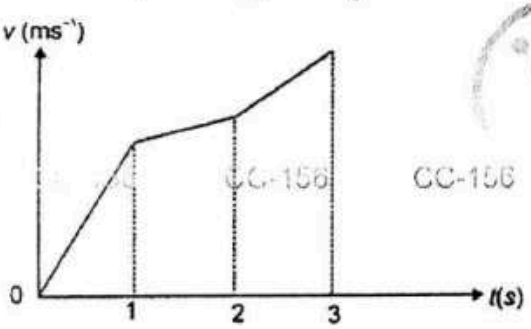
The acceleration-time (a-t) plot of a motion of a body moving along a straight line is shown below. If the particle starts from rest, then choose the correct (v-t) velocity-time

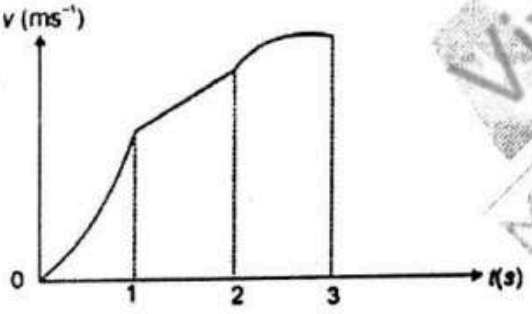
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graph among the options given below.

(1)


(2)


(3)


(4)


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45. Two bodies of masses $4M$ and $9M$ are placed at a distance r apart. The gravitational potential at a point on the line joining them where gravitational field is zero, is

(1) $-\frac{20GM}{r}$

(2) $-\frac{25GM}{r}$

(3) $-\frac{15GM}{r}$

(4) $-\frac{30GM}{r}$

CHEMISTRY

46. Match list I with list II

	List I (Species)		List II (Hybridisation and shape)
a.	SF_4	(i)	sp^3d and T-shape
b.	BrF_5	(ii)	sp^3d^2 and square planar
c.	XeF_4	(iii)	sp^3d and see saw
d.	ClF_3	(iv)	sp^3d^2 and square pyramidal

Choose the correct option.

(1) a(i), b(iv), c(ii), d(iii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(i), b(ii), c(iv), d(iii)

(4) a(iii), b(iv), c(i), d(ii)

47. The de-Broglie wavelength of a particle with velocity 5×10^3 m/s is (mass of particle = 0.33 kg, $h = 6.6 \times 10^{-34}$ Js)

(1) 4×10^{-36} m

(2) 2×10^{-35} m

(3) 2×10^{-37} m

(4) 4×10^{-37} m

48. 20 moles of a gas compressed reversibly from 100 L to 10 L at constant temperature. The entropy change during the process is

(1) –23.03 R

(2) 46.06 R

(3) –46.06 R

(4) zero

49. The solubility of $Pb(OH)_2$ in water is 4×10^{-6} M, then solubility of $Pb(OH)_2$ in the buffer solution of pH = 9 is

(1) 2.56×10^{-6} M

(2) 1.25×10^{-4} M

(3) 2.45×10^{-3} M

(4) 5.6×10^{-8} M

50. During nitrogen estimation of an organic compound by Kjeldahl's method, the ammonia evolved by 0.5 g of the compound neutralise 5 mL of 1 M H_2SO_4. The percentage of nitrogen in the compound is

(1) 42%

(2) 56%

(3) 14%

(4) 28%

51. Consider the following two statements.
Statement I: Orthophosphorus acid on heating give orthophosphoric acid and phosphine.
Statement II: Orthophosphorus acid contains one P=O and 3 P–OH bonds.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

52. Elevation in boiling point of an aqueous solution of 0.1 molal monobasic acid, which undergoes 25% ionization is ($K_b(H_2O) = 0.52$ K molal^{-1})

(1) 0.2 K

(2) 0.065 K

(3) 0.13 K

(4) 0.104 K

53. E_{cell} of the cell $Mg|Mg^{2+}(0.1\ M) \parallel Ag^+(0.001\ M) | Ag$ at 25°C is nearly $(E^\circ_{cell} = 3.17\ V)$

(1) 1.5 V

(2) 3 V

(3) 3.5 V

(4) 4 V

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54. Match list I with list II

	List I (Species)		List II (Oxidation state of central atom)
a.	C_3O_2	(i)	+7
b.	Br_3O_8	(ii)	+5
c.	$HClO_4$	(iii)	0
d.	H_3PO_4	(iv)	+4

Choose the correct option

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

55. Which of the following has most negative heat of hydration?

(1) Cu^{2+}

(2) Fe^{2+}

(3) Cr^{2+}

(4) Ti^{2+}

56. Given below are two statements

Statement I: Ethanal is miscible with water in all proportions.

Statement II: Ethanal forms hydrogen bond with water.

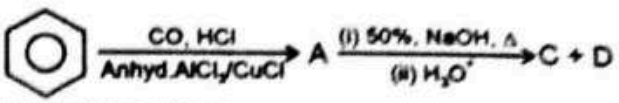
In light of above statements, choose the **correct** option.

(1) Statement I is incorrect but statement II is correct

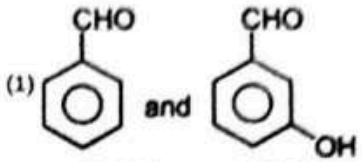
(2) Both statement I and statement II are incorrect

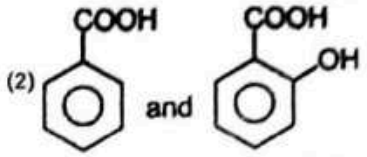
(3) Both statement I and statement II are correct

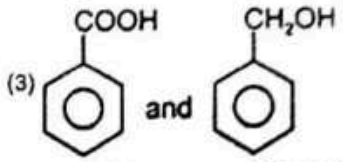
(4) Statement I is correct but statement II is incorrect

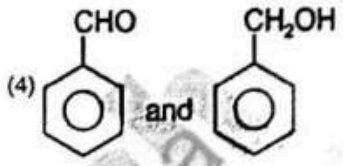
57. 

Products C and D are

(1) 

(2) 

(3) 

(4) 

58. Match groups in list-1 with their group reagent in list-2 and choose **correct** answer.

List-1		List-2	
(a)	Group-I	(i)	NH_4OH in presence of NH_4Cl
(b)	Group-II	(ii)	H_2S in presence of NH_4OH
(c)	Group-III	(iii)	Dilute HCl
(d)	Group-IV	(iv)	H_2S gas in presence of dil. HCl

(1) (a)(iii), (b)(iv), (c)(i), (d)(ii)

(2) (a)(i), (b)(ii), (c)(iii), (d)(iv)

(3) (a)(iv), (b)(iii), (c)(ii), (d)(i)

(4) (a)(ii), (b)(i), (c)(iv), (d)(iii)

59. When dimethyl glyoxime is added to the aqueous solution of nickel chloride in an alkaline medium, it forms a complex having

(1) Black colour

(2) Red colour

(3) Blue colour

(4) Yellow colour

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60. Which among the following method can be used for the preparation of benzene?
(i) By passing ethyne through red hot iron tube
(ii) On heating sodium benzoate with soda lime
(iii) By passing vapours of phenol over heated zinc dust

(1) (i) and (ii) only

(2) (ii) and (iii) only

(3) (i) and (iii) only

(4) (i), (ii) and (iii)

61. If haemoglobin contains 0.334% iron by mass and one molecule of haemoglobin has 4 atoms of iron then the molecular mass of haemoglobin is approximately

(1) 98432.4 u

(2) 54298.6 u

(3) 24502.5 u

(4) 67065.8 u

62. Consider the following statements
(a) Dipole moment of H_2O is greater than the NH_3
(b) C≡C bond length is smaller than the C≡N bond length
(c) Both CO and NO^+ have bond order 2
(d) The formal change on central oxygen atom in ozone molecule is +2
Choose the incorrect statements

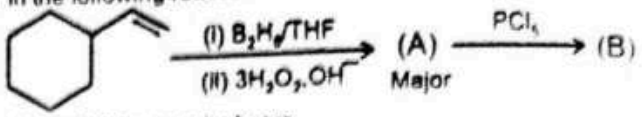
(1) (a) and (c) only

(2) (b) and (c) only

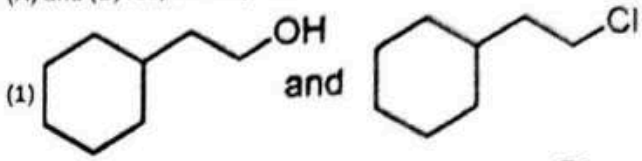
(3) (b), (c) and (d) only

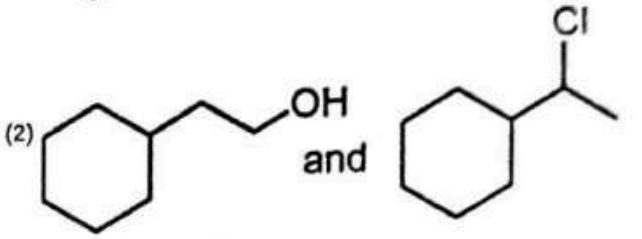
(4) (a) and (d) only

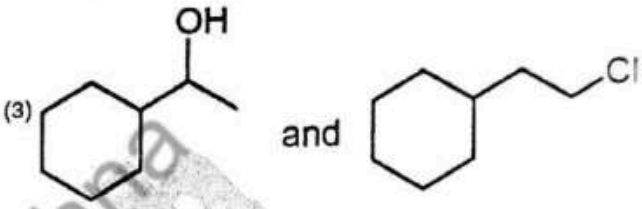
63. In the following reaction

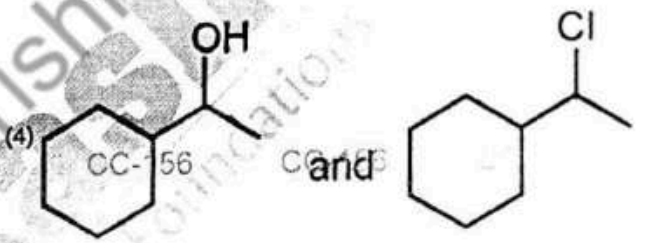


(A) and (B) respectively are

(1) 

(2) 

(3) 

(4) 

64. An organic compound (X), molecular formula (C_6H_{12}) on reductive ozonolysis gives only two moles of compound (Y). Compound (Y) does not give Fehling's test. It undergoes aldol condensation and gives compound (Z). The compound (Z) will be

(1) 3-Methylpent-3-en-2-one

(2) 3-Methylpent-3-one

(3) 4-Methylpent-3-en-2-one

(4) 4-Methyl-but-2-enal

65. Mechanism of a hypothetical reaction $2A_3 \rightarrow 3A_2$ is given below
(i) $A_3 \rightleftharpoons A_2 + A$ (fast)
(ii) $A_3 + A \rightarrow 2A_2$ (slow)
The overall order of the reaction will be

(1) 0

(2) 3

(3) 1

(4) 2

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66. Rate of formation of NH_3 according to the reaction, $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ is 4×10^4 mol L^{-1} s^{-1}, then the rate of disappearance of H_2 will be

(1) 1.4×10^4

(2) 6×10^4

(3) 2.7×10^4

(4) 12×10^4

67. Consider the following two statements.
Statement I: Amylose is water soluble component which constitutes about 80-85% of starch.
Statement II: Amylopectin is water insoluble component which constitutes about 15-20% of starch.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

68. Choose the incorrect statement

(1) DNA molecule contains β-D-2-deoxyribose sugar moiety

(2) RNA molecule contains β-D-ribose sugar moiety

(3) DNA contains adenine, guanine, cytosine and uracil bases

(4) RNA contains adenine, guanine, cytosine and uracil bases

69. Choose the incorrect match.

	Order		Property
(a)	$H_2O > H_2Te > H_2Se > H_2S$	–	Melting point
(b)	$H_2Te > H_2Se > H_2S > H_2O$	–	Dissociation constant
(c)	$H_2O > H_2S > H_2Se > H_2Te$	–	Bond angle
(d)	$H_2Te > H_2Se > H_2S > H_2O$	–	Boiling point

(1) (a)

(2) (b)

(3) (c)

(4) (d)

70. The correct order of negative electron gain enthalpies is

(1) Se > O > S > Te > Po

(2) O > S > Se > Te > Po

(3) S > Se > Te > Po > O

(4) Po > Te > Se > S > O

71. Match list I with list II

	List I (Species)		List II (Nature)
a.	GeO_2	(i)	Basic
b.	CO	(ii)	Acidic
c.	Tl_2O_3	(iii)	Amphoteric
d.	Ga_2O_3	(iv)	Neutral

Choose teh correct option

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(ii), b(iii), c(i), d(iv)

(4) a(i), b(iv), c(iii), d(ii)

72. The correct electronic configuration of gadolinium and lutetium respectively are

(1) $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^{14} 6s^2$

(2) $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^{14} 5d^1 6s^2$

(3) $[Xe]4f^8 6s^2$ and $[Xe]4f^{14} 6s^2$

(4) $[Xe]4f^8 6s^2$ and $[Xe]4f^{14} 5d^1 6s^2$

73. Consider the following statements about $[Co(NH_3)_6]^{3+}$ complex ion.
(a) Central atom is d^2sp^3 hybridised
(b) It is dimagnetic in nature
(c) It is spin free complex ion
Choose the correct statements.

(1) (a) and (b) only

(2) (a), (b) and (c)

(3) (a) and (c) only

(4) (b) and (c) only

74. The correct IUPAC name of $[PtCl_2(en)_2](NO_3)_2$ is

(1) Bis(ethane-1,2-diamine)dichloridoplatinum(III) nitrate

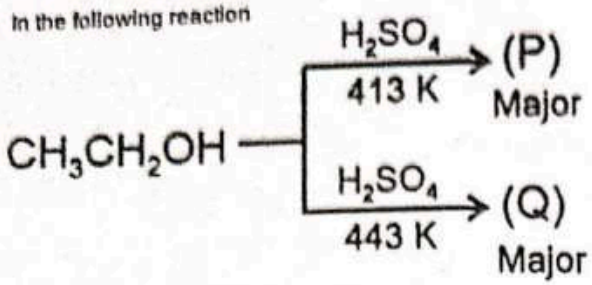
(2) Ethane-1,2-diaminedichloridoplatinum(III) nitrate

(3) Dichloridobis(ethane-1,2-diamine)platinum(IV) nitrate

(4) Bis(ethane-1,2-diamine)dichloridoplatinum(IV) nitrate

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75. In the following reaction



Products (P) and (Q) respectively are

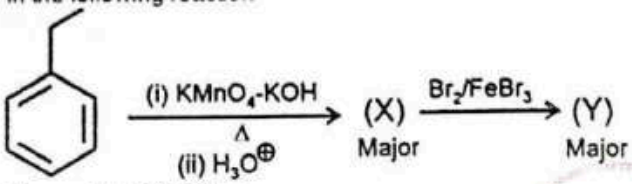
(1) $C_2H_5OC_2H_5$ and $CH_2=CH_2$

(2) CH_3CHO and $CH_2=CH_2$

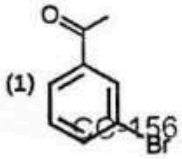
(3) $CH_2=CH_2$ and CH_3CHO

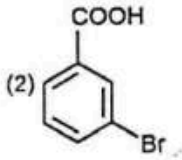
(4) $C_2H_5OC_2H_5$ and CH_3COOH

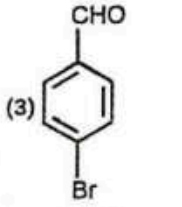
76. In the following reaction

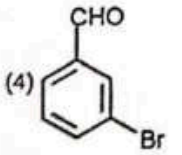


The product (Y) will be

(1) 

(2) 

(3) 

(4) 

77. The correct order of acidic strength of the following compound is

(1) $ClCH_2CH_2COOH > HCOOH > C_6H_5COOH > NC-CH_2-COOH$

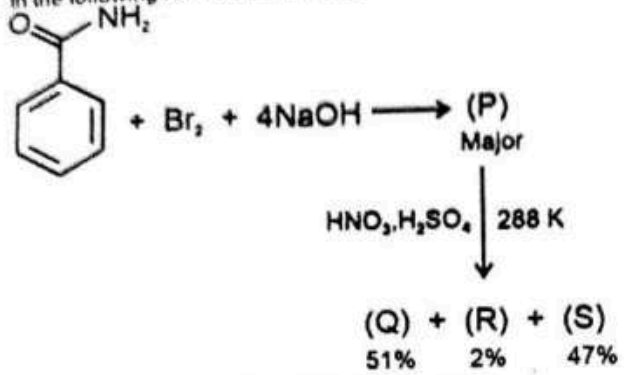
(2) $C_6H_5COOH > NC-CH_2-COOH > HCOOH > ClCH_2CH_2COOH$

(3) $HCOOH > NC-CH_2-COOH > C_6H_5COOH > ClCH_2CH_2COOH$

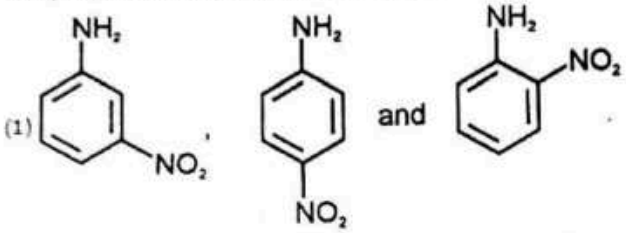
(4) $NC-CH_2-COOH > HCOOH > ClCH_2CH_2COOH > C_6H_5COOH$

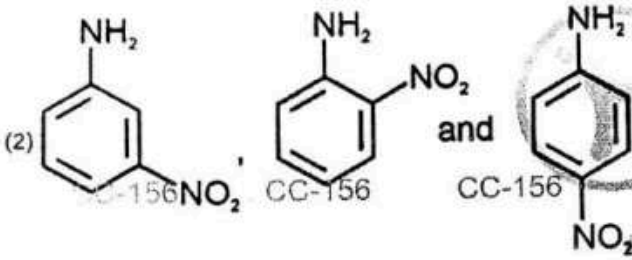
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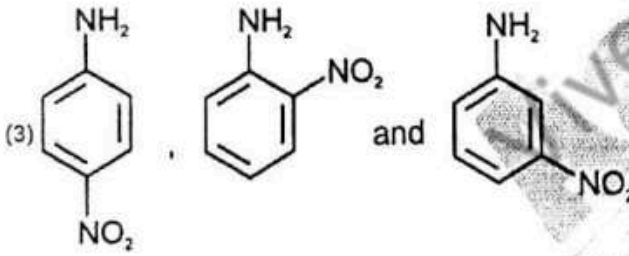
78. In the following reaction sequences

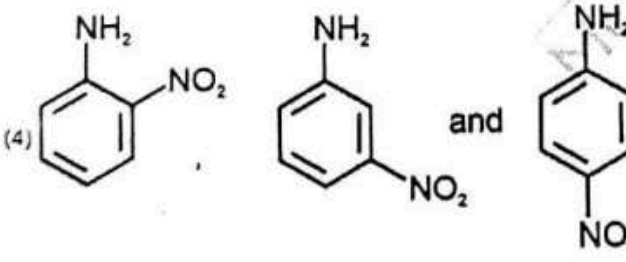


The product (Q), (R) and (S) respectively are

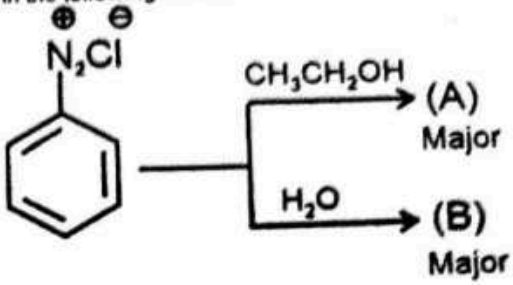
(1) 

(2) 

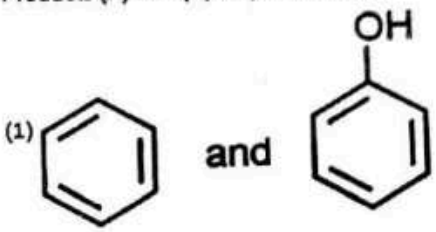
(3) 

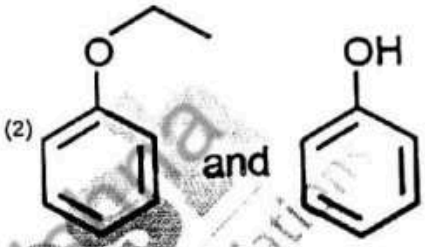
(4) 

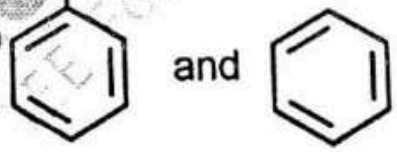
79. In the following reaction

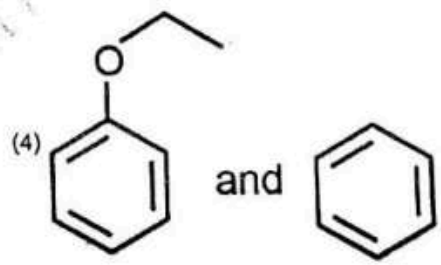


Products (A) and (B) respectively are

(1) 

(2) 

(3) 

(4) 

CC-156

80. Given below are two statements, one is labelled as assertion (A) and other is labelled as reason (R).
Assertion (A): Aromatic primary amines cannot be prepared by the Gabriel phthalimide synthesis.
Reason (R): In Gabriel phthalimide synthesis, aryl halide does not undergo nucleophilic substitution with the anion formed by phthalimide.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are false

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) (A) is true but (R) is false

81. Mass of 80% pure $CaCO_3$ required to neutralise 100 mL of 0.1 M HCl solution according to following reaction is
$CaCO_3(s) + 2HCl \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

(1) 0.505 g

(2) 0.625 g

(3) 0.745 g

(4) 0.895 g

82. Consider the following two statements.
Statement I: Both N_2^+ and N_2^- have same bond order.
Statement II: Both N_2^+ and N_2^- are paramagnetic in nature.
In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

83. $C(s) + O_2(g) \rightarrow CO_2(g) + X$ kJ
$2CO(g) + O_2(g) \rightarrow 2CO_2(g) + Y$ kJ
The enthalpy of formation of CO(g) (in kJ) is

(1) X – Y

(2) 2X – Y

(3) $X - \frac{Y}{2}$

(4) $\frac{Y}{2} - X$

84. If the pH of ammonium acetate solution is 7.005 and pK_a of acetic acid is 4.76 then the pK_b of ammonium hydroxide is

(1) 4.82

(2) 4.24

(3) 4.38

(4) 4.75

85. If 0.04 M solution of Na_2SO_4 (100% ionization) is isotonic with aqueous solution of glucose at 27°C, then molarity of aqueous solution of glucose is

(1) 0.2 M

(2) 0.3 M

(3) 0.12 M

(4) 0.5 M

86. Consider the following statements.
(a) Thin layer chromatography is an example of adsorption chromatography
(b) Silica gel and alumina are common adsorbents used in thin layer chromatography
(c) Paper chromatography is a type of partition chromatography
The correct statement(s) is/are

(1) (a) only

(2) (c) only

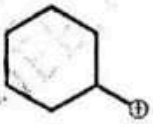
(3) (a) and (c) only

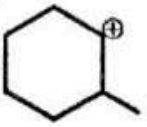
(4) (a), (b) and (c)

87. Which of the following is the most stable carbocation?

(1) 

(2) 

(3) 

(4) 

88. Match the given Lists-I and List-II choose the correct option:

List-I Vitamin		List-II Deficiency diseases	
a.	Vitamin B_{12}	(i)	Scurvy
b.	Vitamin C	(ii)	Convulsions
c.	Vitamin D	(iii)	Pernicious anaemia
d.	Vitamin B_6	(iv)	Rickets

(1) a(iii), b(i), c(iv), d(ii)

(2) a(i), b(iii), c(iv), d(ii)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iii), c(ii), d(iv)

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89. Given below are two statements, one is labelled as **Assertion (A)** other is labelled as **Reason (R)**.

Assertion(A) : Boron cannot form $[BF_6]^{3-}$.

Reason(R) : Maximum covalency of B is 4 due to the absence of vacant d-orbitals.

In the light of above statements, choose the correct answer.?

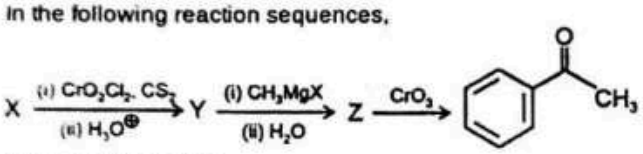
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

90. In the following reaction sequences,



The compound X is

(1) Benzaldehyde

(2) Benzene

(3) Toluene

(4) Phenol

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BOTANY

91. In a pond, there were 40 water lily plants last year. Calculate the birth rate if their population reaches to 56 by reproduction in one year and choose the **correct** answer from the options given below.

(1) 0.4 individuals per water lily per year

(2) 0.8 individuals per water lily per year

(3) 1.6 individuals per water lily per year

(4) 16 individuals per water lily per year

92. Find the **odd one** out w.r.t. cytokinin.

(1) It promotes cell division

(2) It initiates the growth of lateral buds

(3) Delay the senescence of leaves

(4) Derived from acetyl-CoA

93. In post-transcriptional events, splicing is a process in which

(1) Exons are removed from primary transcript

(2) Introns are removed and exons are joined in a defined order

(3) Adenylate residues are added at 3' end of hnRNA

(4) Methyl guanosine triphosphate is added at 5' end of mature mRNA

94. Match List I with List II.

List I	List II
(a) Atmospheric N_2-fixer	(i) *Glomus*
(b) Mycorrhiza	(ii) *Azospirillum*
(c) Baculoviruses	(iii) *Nucleopolyhedrovirus*
(d) Free-living fungus	(iv) *Trichoderma*

Choose the **correct** option.

(1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(4) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

95. Conjoint and closed vascular bundles are commonly found in

(1) Dicot roots

(2) Leaves of flowering plants

(3) Dicot stems

(4) Monocot roots

96. Select the **mismatched** pair w.r.t. human

(1) Kingdom – Animalia

(2) Order – Carnivora

(3) Genus – *Homo*

(4) Phylum – Chordata

97. Mark the **incorrect** one w.r.t. human genome

(1) Less than 2% of genome codes for proteins

(2) The average gene consists of 3000 bases

(3) Repeated sequences make up a small portion of human genome

(4) The functions of over 50% of discovered genes are unknown

98. Which of the following is **not** a dominant trait of the pea plant according to Mendel's studies?

(1) Constricted pod shape

(2) Violet flower colour

(3) Round seed

(4) Axial flower position

99. The sub-stage of prophase I which is followed by the tendency of the recombined homologous chromosomes of the bivalents to separate from each other is

(1) Zygotene

(2) Diplotene

(3) Pachytene

(4) Leptotene

100. An angiospermic family which includes a plant that produces colchicine

(1) Bears non endospermous seeds

(2) Has tricarpellary superior ovary

(3) Has flowers which show zygomorphic symmetry

(4) Has unisexual flowers only

101. Sexual deceit to get the pollination done is employed by the plant

(1) *Ophrys*

(2) Fig

(3) Mango

(4) *Calotropis*

102. Adventitious roots are seen in

(1) Mustard

(2) *Monstera*

(3) Carrot

(4) Turnip

103. Which of the following statements is true?

(1) Robert May estimated the global species diversity to be 20-50 million.

(2) Among animals, insects are the most species-rich taxonomic group.

(3) More than 70% of all the species recorded are plants.

(4) For many taxonomic groups, species inventories are more complete in tropical than in temperate countries.

104. Each bivalent in pachytene stage shows

(1) 2 chromosomes, 2 chromatids

(2) 4 chromosomes, 2 chromatids

(3) 2 chromosomes, 4 chromatids

(4) 4 chromosomes, 4 chromatids

105. Which of the following can be placed under the class Ascomycetes?

(1) *Claviceps*

(2) *Ustilago*

(3) *Agaricus*

(4) *Albugo*

106. At least how many meiotic divisions are required to produce 100 seeds in a typical flowering plant?

(1) 100

(2) 25

(3) 125

(4) 50

107. Some cells of shoot apical meristem that are left behind in axil of leaves

a. Constitute axillary buds
b. Are secondary meristem
c. Are capable of forming a flower
d. Form intercalary meristem

The **correct** one(s) is/are

(1) a and d

(2) a only

(3) b and c

(4) a and c

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108. Cyclic photophosphorylation

(1) is connected with photolysis of water

(2) Synthesises both ATP as well as NADPH

(3) is performed by photosystem I independently

(4) Requires an external source of electron donor

109. Read the following statements of Assertion (A) and Reason (R) and choose the correct option.
Assertion (A): Down's syndrome is said to be a chromosomal disorder.
Reason (R): In an individual, Down's syndrome is the result of an extra copy of chromosome 21.

(1) Both (A) and (R) are false

(2) (A) is true but (R) is false

(3) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

110. The term 'nuclein' was given by

(1) James Watson

(2) Friedrich Meischer

(3) Rosalind Franklin

(4) Erwin Chargaff

111. In human beings, the membrane of erythrocyte has approximately

(1) 42% proteins and 50% lipids

(2) 52% proteins and 40% lipids

(3) 40% proteins and 52% lipids

(4) 60% proteins and 2% lipids

112. If, in a food chain, 15 J of energy is trapped at producer level, then how much energy will be available to tertiary consumer (T_4 level)?

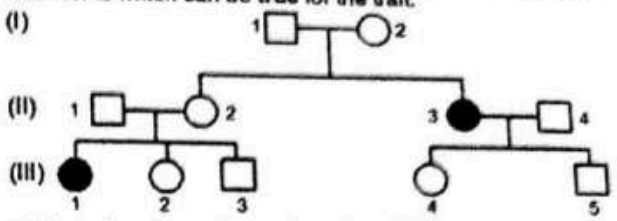
(1) 0.015 J

(2) 0.0015 J

(3) 0.15 J

(4) 0.00015 J

113. Study the given pedigree chart of a family and select the statements which can be true for the trait.



a. The trait under study can be colour blindness.
b. The genotype of II-2 cannot be AA.
c. The pedigree can be for inheritance of cystic fibrosis.
d. The genotype of III-4 can be AA.

(1) a and b

(2) b and c only

(3) a and c

(4) b, c and d

114. In the equation, log S = log C + Z log A, representing the species-area relationship, Z stands for:

(1) Species richness

(2) Slope of the line

(3) Y-intercept

(4) Area

115. Filiform apparatus

(1) Is present in generative cell of pollen

(2) Guides the entry of pollen tube into synergid

(3) Is composed of sporopollenin

(4) Absorbs the water for seed germination

116. Choose the **incorrect** match among the following

(1) Marginal placentation – Pea

(2) Parietal placentation – Lemon

(3) Free central placentation – *Dianthus*

(4) Basal placentation – Sunflower

117. Match the following complexes to their alternate name and select the **correct** option.

	Column I		Column II
a.	Complex I	(i)	Cytochrome bc_1 complex
b.	Complex II	(ii)	Cytochrome *c* oxidase complex
c.	Complex III	(iii)	NADH dehydrogenase
d.	Complex IV	(iv)	Succinate dehydrogenase

(1) a(iii), b(iv), c(i), d(ii)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iii), b(i), c(ii), d(iv)

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118. Anisogamous reproduction can be observed in

(1) *Volvox*

(2) *Fucus*

(3) *Eudorina*

(4) *Ulothrix*

119. How many of the following structures in a cell are not bounded by any membrane?
Lysosome, Vacuole, Ribosome, Centrosome, Cilium, Nucleolus

(1) Two

(2) Three

(3) Five

(4) Four

120. Select the incorrect statements from the following.
(a) Hatch and Slack pathway occurs in mesophyll cells of all types of plants.
(b) ATP required for phosphorylation during reduction stage in Calvin cycle comes from mitochondria.
(c) In *Sorghum*, the primary CO_2 acceptor is a 3-C molecule.
(d) C_4 plants lack photorespiration.

(1) (a) & (d) only

(2) (a) & (b) only

(3) (b) & (c) only

(4) (a), (b) & (c) only

121. Which of the following are not seen in a prokaryotic cell?

(1) Ribosomes

(2) Lysosomes

(3) Mesosomes

(4) Plasmids

122. Observe the given tRNA



Select the option that cannot be true regarding the above tRNA?

(1) This tRNA will bind at AUG codon of mRNA

(2) This tRNA binds with amino acid methionine at its 3' end

(3) The loop with UAC is anticodon loop of this tRNA

(4) The amino acid X would be the same when anticodon UAC is changed to UAA

123. In flowering plants, microsporangium develops to form

(1) Embryo sac

(2) Pollen sac

(3) Pollen grain

(4) Ovule

124. Which of the following is not the basis of classification of fungi?

(1) Morphology of the mycelium

(2) Mode of spore formation

(3) Mode of nutrition

(4) Fruiting bodies

125. A couple has two daughters and two sons. Among two daughters one is homozygous normal and other is carrier for colour blindness, whereas, among two sons one is normal and other is colour blind. What should be genotypes of the parents?

(1) XX and XY

(2) XX and X^CY

(3) XX^C and XY

(4) XX^C and X^CY

126. Which of the following statements is false w.r.t. phenylketonuria?

(1) It is an inborn error of metabolism

(2) Along with accumulation of the derivatives of phenylalanine these are also excreted through urine

(3) Affected individual lacks phenylalanine hydroxylase enzyme

(4) Phenylalanine is converted into tyrosine that when accumulated in brain results in mental retardation

127. Find the odd one w.r.t. brown algae.

(1) Show great variation in size and form

(2) They contain chlorophyll a, chlorophyll c and fucoxanthin

(3) They reproduce asexually only by non-motile spores

(4) Their gametes are pyriform bearing two laterally attached flagella

128. Dinoflagellates differ from euglenoids, as the former

(1) Can predate on other smaller organisms

(2) Have majority of members that are fresh water organisms

(3) Have two flagella

(4) Have cell wall with stiff cellulose plates on the outer surface

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129. Read the following statements and select the correct option.
Statement A: In *lac* operon the promoter gene adjacent to the operator gene interacts with protein molecule synthesized by repressor mRNA.
Statement B: Regulation of *lac* operon by repressor is referred to as positive regulation.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are incorrect

(4) Both the statements A and B are correct

130. What is the net ATP gain when one molecule of glucose is fermented to alcohol or lactic acid?

(1) 6

(2) 2

(3) 38

(4) 8

131. The genetic codes are said to be

(1) Doublet

(2) Degenerate

(3) Absolutely universal

(4) Read in mRNA in discontinuous fashion as there are punctuations

132. Sclerenchyma tissue

(1) Is a living tissue

(2) Consists of narrow cells with thick and lignified cell wall

(3) Has the cells which are oval in shape and often contain chloroplast

(4) Consists of cell with thin and cellulosic cell wall

133. Heterophylly due to difference in environment is seen in

(1) Cotton

(2) Coriander

(3) Larkspur

(4) Buttercup

134. Match the following columns and choose the correct option.

Column I		Column II	
a.	Fluid-mosaic model	(i)	Golgi apparatus
b.	Calcium pectate	(ii)	Plasma membrane
c.	Divide by fission	(iii)	Middle lamella
d.	Formation of glycoproteins and glycolipids	(iv)	Mitochondria

(1) a(i), b(iv), c(ii), d(iii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(i), b(ii), c(iii), d(iv)

135. In which of the following population interactions, only one species gets benefit?
a. Mutualism
b. Predation
c. Parasitism
d. Amensalism
e. Commensalism
The correct ones are

(1) a and b only

(2) b, c and e

(3) c, d and e

(4) a, c and d

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ZOOLOGY

136. Select the correct set of favourable conditions required for the formation of oxyhaemoglobin in the alveoli.

(1) Low pO_2, low pCO_2, more H^+, higher temperature

(2) High pO_2, low pCO_2, less H^+, lower temperature

(3) Low pO_2, high pCO_2, more H^+, higher temperature

(4) High pO_2, high pCO_2, less H^+, higher temperature

137. Observe the reactions given below.

Prothrombin → Thrombin

Fibrinogens → Fibrins

Identify A and B and choose the correct option.

(1) A - Thrombokinase; B - Heparin

(2) A - Thrombins; B - Thrombokinase

(3) A - Thrombokinase; B - Thrombin

(4) A - Tissue factor; B - Platelet factor

138. How many biomolecules given in the box below are found in the acid-soluble fraction obtained upon chemical analysis of a living tissue?

Glucose, Ribose, Lipids, Glutamic acid, Collagen, Starch

Choose the correct option.

(1) Two

(2) Three

(3) Four

(4) Five

139. Which of the following co-factors forms bond with the enzyme carbonic anhydrase?

(1) Zinc

(2) Haem

(3) Copper

(4) Iron

140. A joins skeletal muscle to bone whereas B connects one bone to another. Here A and B are

(1) A – Areolar tissue, B – Ligament

(2) A – Tendon, B – Areolar tissue

(3) A – Tendon, B – Ligament

(4) A – Ligament, B – Tendon

141. Select the pair of hormones synthesised by the hypothalamus and transported axonally to the pituitary gland.

(1) Prolactin and TSH

(2) Oxytocin and vasopressin

(3) FSH and LH

(4) MSH and prolactin

142. Both *Scoliodon* and *Betta* are true fishes, but they differ from each other as

(1) Former has operculum whereas latter does not

(2) Only former has a streamlined body

(3) Former lives in marine habitat while latter is an aquarium fresh water fish

(4) Fertilisation in former is external while latter exhibits internal fertilisation

143. Substances like glucose, amino acids, Na^+, *etc.*, in the renal filtrate are reabsorbed ____ whereas the nitrogenous wastes are absorbed by ____ transport. Reabsorption of water also occurs ____ in the initial segments of nephron.

Fill the blanks **correctly** from given options.

	a	b	c
(1)	Actively	Passive	Passively
(2)	Passively	Active	Passively
(3)	Actively	Passive	Actively
(4)	Passively	Passive	Actively

(1) (1)

(2) (2)

(3) (3)

(4) (4)

are

(1) G-actin and meromyosin

(2) Tropomyosin and troponin

(3) Meromyosin and G-actin

(4) G-actin and troponin

145. Conditional reabsorption of water occurs under the influence of ADH by which of the following parts of nephron?

(1) Loop of Henle

(2) PCT

(3) DCT

(4) Glomerulus

146. What is true about *Balanoglossus*, *Sepia* and *Antedon*?

(1) All belong to the same phylum

(2) All possess water vascular system

(3) All exhibit radial symmetry

(4) All possess true coelom

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147. The first pair of wings of cockroach arises from A and is known as B.
Select the option which fills A and B correctly.

	A	B
(1)	Metathorax	Metathoracic wings
(2)	Mesothorax	Tegmina
(3)	Mesothorax	Metathoracic wings
(4)	Metathorax	Tegmina

(1) (1)

(2) (2)

(3) (3)

(4) (4)

148. In frogs, the medulla oblongata passes out through the _____ and continues into spinal cord.
Select the option to fill the blank correctly.

(1) Olfactory lobe

(2) Foramen magnum

(3) Diencephalon

(4) Optic lobe

149. The outermost protective covering of the central information processing organ of our body is

(1) Dura mater

(2) Arachnoid mater

(3) Pia mater

(4) CSF

150. Read the following statements and choose the **correct** option.
Statement A: Muscle contraction in smooth muscles can be explained by the sliding filament theory.
Statement B: Relaxation of skeletal muscle fibres takes place when thick filaments slide over thin filaments.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

151. Choose the **incorrect** option w.r.t. mode of action of steroidal oral contraceptive pills.

(1) Inhibit the release of ovum from ovary

(2) Alter the quality of cervical mucus

(3) Prevent the embedding of blastocyst in uterine endometrium

(4) Release copper ions to suppress sperm motility

152. Complete the analogy w.r.t. ploidy level and select the correct option.
Diploid : Spermatogonium :: Haploid : _____

(1) Primary spermatocyte

(2) Primary oocyte

(3) Secondary oocyte

(4) Oogonium

153. All of the following features are correct w.r.t. a plasmid, **except**

(1) Autonomously replicating

(2) Circular in shape

(3) Acts as the chromosomal genetic material

(4) Acts as an extra-chromosomal material

154. The hormone which attains the peak twice during the menstrual cycle in a normal human female is

(1) LH

(2) FSH

(3) Estrogen

(4) Progesterone

155. Read the given description carefully
There are four DNA fragments: A, B, C and D. Given, C = B + A, B < A, and D is the smallest of all.
Which of the following represents correct sequence of bands from cathode to anode when these segments are separated with the help of gel electrophoresis?

(1) C, B, A, D

(2) C, A, D, B

(3) C, A, B, D

(4) D, B, A, C

156. Which of the following cannot act as the source of complementary RNA during RNAi?

(1) Transposons that replicate *via* an RNA intermediate

(2) Infection by viruses having RNA genomes

(3) Mobile genetic elements that replicate *via* an RNA intermediate

(4) Natural *Agrobacterium tumefaciens*

157. Bacteria like *Streptococcus pneumoniae* and _____ are responsible for the disease pneumonia in humans.
Fill in the blank by selecting the correct option.

(1) *Salmonella typhi*

(2) *Haemophilus influenzae*

(3) *Epidermophyton*

(4) *Bacillus thuringiensis*

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158. Arrange the given geological periods in the correct sequence from oldest to most recent.
(a) Triassic
(b) Permian
(c) Devonian
(d) Jurassic
(e) Cretaceous
Select the correct option.

(1) (a) → (b) → (d) → (e) → (c)

(2) (c) → (b) → (a) → (d) → (e)

(3) (c) → (e) → (b) → (d) → (a)

(4) (d) → (a) → (c) → (e) → (b)

159. Select the **incorrect** match.

(1) Origin of the first non-cellular form of life - 3 bya

(2) Origin of the Universe - 4 bya

(3) Origin of the Earth - 4.5 bya

(4) Origin of the first cellular form of life - 2000 mya

160. In uterus of a human female, the _____ layer is made up of smooth muscles, while _____ layer is glandular in nature.
Select the correct option to fill in the respective blanks.

(1) Endometrium, myometrium

(2) Myometrium, endometrium

(3) Perimetrium, endometrium

(4) Endometrium, perimetrium

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161. An example of analogous structures among the following is

(1) Thorns of *Bougainvillea* and tendrils of *Cucurbita*

(2) Hearts of frog and humans

(3) Forelimbs of mammals and bats

(4) Eyes of *Octopus* and mammals

162. Select the **incorrect** option w.r.t. vaccines.

(1) The principle of immunisation through vaccination is based on creation of immunological memory

(2) In vaccination, antigenic proteins or weakened pathogens are introduced in the body, providing artificially acquired passive immunity

(3) Recombinant DNA technology can allow the production of antigenic polypeptides of pathogens in bacteria or yeast for preparation of some vaccines

(4) Recombinant hepatitis-B vaccine has been produced from yeast

163. In a population of 900 cows, which are in Hardy-Weinberg equilibrium, 891 are white and 9 are black. What would be frequency of dominant and recessive allele respectively within this population?

(1) 0.8 and 0.2

(2) 0.09 and 0.01

(3) 0.9 and 0.1

(4) 0.08 and 0.02

164. Consider the following statements w.r.t. evolution and select the **correct** option.
Statement (A): Genetic variability is the raw material for operation of natural selection.
Statement (B): Adaptive ability is inherited and has a genetic basis.

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both statements (A) and (B) are correct

(4) Both statements (A) and (B) are incorrect

165. Choose the **incorrect** statement w.r.t. human sperms.

(1) The neck of sperm is filled with hydrolytic enzymes that help in fertilisation of the ovum.

(2) Male ejaculates about 200-300 million sperms during a coitus.

(3) The middle piece of sperm possesses numerous mitochondria.

(4) Acrosome is anteriorly located to the nucleus and contains enzymes that help in fertilisation of ovum.

166. At resting state *i.e.*, when a neuron is not conducting any impulse, the axonal membrane is

(1) Completely impermeable to Na^+

(2) Completely impermeable to K^+

(3) Nearly impermeable to Na^+

(4) Only permeable to negatively charged proteins

167. Consider the given statements
Statement A: Purified DNA precipitates out after addition of chilled ethanol and can be seen as fine threads in the suspension.
Statement B: In the method of micro-injection, recombinant DNA is directly injected into mitochondria of a plant cell.
Select the **correct** option.

(1) Only statement A is correct

(2) Both statements A and B are correct

(3) Both statements A and B are incorrect

(4) Only statement B is correct

168. Recovery of healthy plants from diseased plants is possible by

(1) Meristem culture

(2) Somatic hybridisation

(3) Protoplast fusion

(4) Bioprocess engineering

169. Which part of the brain has special neural centre that can moderate the cardiac function through autonomic nervous system (ANS)?

(1) Cerebrum

(2) Cerebellum

(3) Medulla oblongata

(4) Corpora quadrigemina

170. How many fragments will be obtained if we digest cloning vector pBR322 with restriction enzymes *Cla* I and *Sal* I?

(1) 3

(2) 1

(3) 4

(4) 2

171. Match the Column I with Column II w.r.t. humans and choose the **correct** option.

	Column I		Column II
A.	Ball and socket joint	i.	Between carpals
B.	Hinge joint	ii.	Between carpal and metacarpal of thumb
C.	Saddle joint	iii.	Shoulder joint
D.	Gliding joint	iv.	Knee joint

(1) A-ii, B-i, C-iv, D-iii

(2) A-iii, B-iv, C-ii, D-i

(3) A-iv, B-iii, C-i, D-ii

(4) A-iii, B-iv, C-i, D-ii

172. In humans, which of the following are mainly responsible for maintaining the increasing osmolarity gradient in the medullary interstitium of kidney?

(1) H_2O and NaCl

(2) H_2O and urea

(3) NaCl and K^+

(4) NaCl and urea

173. All of the following sets of hormones can increase the blood glucose levels, **except**

(1) Adrenaline and cortisol

(2) Glucagon and Epinephrine

(3) Glucagon and Cortisol

(4) ACTH and Insulin

174. If an enzyme shows a reaction rate '6X' at a temperature of 50°C, then what will be the rate of this reaction if temperature of the reaction set-up is reduced by 20°C?

(1) 12 X

(2) 3 X

(3) 1.5 X

(4) 8 X

175. Which of the following are the secretions of largest gland of human body?

(1) Hydrocarbons and steroids

(2) NaCl and lactic acid

(3) Lysozyme and ammonia

(4) Bilirubin and biliverdin

176. **Assertion (A):** Ribozymes are proteinaceous enzymes and their catalytic power get affected at high temperature.
Reason (R): The activity of proteinaceous enzyme can be affected by a change in the conditions which can alter the tertiary confirmation of the proteins that form the active sites in the enzyme.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) (A) is false but (R) is true.

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

177. How many animals given in the box below are homeotherms?

Aptenodytes, *Calotes*, *Clarias*, *Myxine*, *Pavo*, *Canis*, *Delphinus*

Select the **correct** option.

(1) Five

(2) Four

(3) Three

(4) Six

178. Read the given statements:
(a) In a healthy individual, under normal conditions, the body is able to differentiate between 'self' and 'non-self' cells.
(b) Blood group matching is essential before undertaking any graft.
(c) After performing successful transplantation, the patient does not require immunosuppressive drugs.
(d) Humoral immune response is primarily responsible for graft rejection.
How many of the above statement(s) is/are **correct** w.r.t. humans?

(1) One

(2) Three

(3) Two

(4) Four

179. Which of the following structures in human female is characterised by the presence of antrum?

(1) Primary follicle

(2) Secondary oocyte

(3) Tertiary follicle

(4) Corpus luteum

180. In a lactating human female, during the period of intense lactation, generally

(1) Both ovulation and menstruation occur normally

(2) Ovulation occurs but menstruation does not occur

(3) Menstruation occurs without ovulation

(4) Both ovulation and menstruation do not occur

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